# Computational Imaging Without a Computer: Seeing Through Random Diffusers at the Speed of Light

**Authors:** Yi Luo[1,2,3], Yifan Zhao[1,3], Jingxi Li[1,2,3], Ege Çetintaş[1,2,3], Yair Rivenson[1,2,3], Mona Jarrahi[1,3], Aydogan Ozcan[1,2,3,*]

[1] Electrical and Computer Engineering Department, University of California, Los Angeles; Los Angeles, CA, 90095, USA

[2] Bioengineering Department, University of California, Los Angeles; Los Angeles, CA, 90095, USA

[3] California NanoSystems Institute, University of California, Los Angeles; Los Angeles, CA, 90095, USA

* Corresponding author. Email: ozcan@ucla.edu

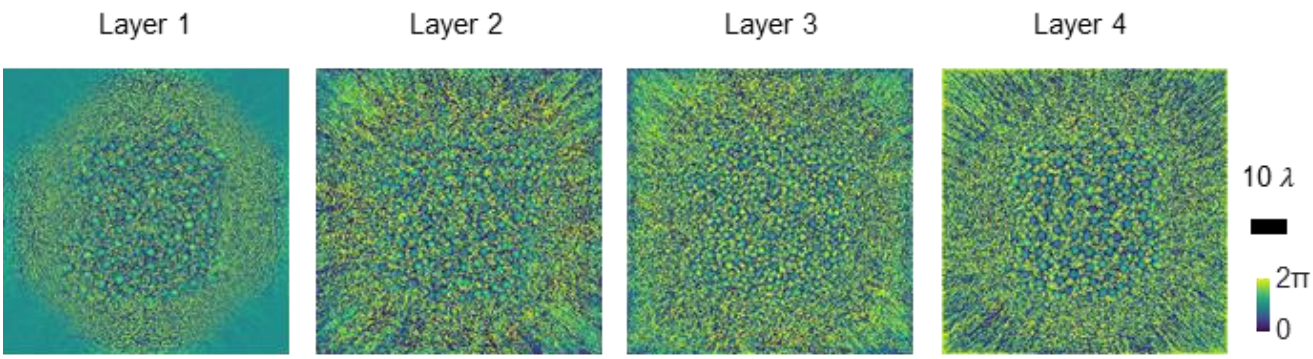


**Figure S1.** Phase patterns of the transmissive layers corresponding to the diffractive network that was trained using *n*=20 diffusers at each epoch.

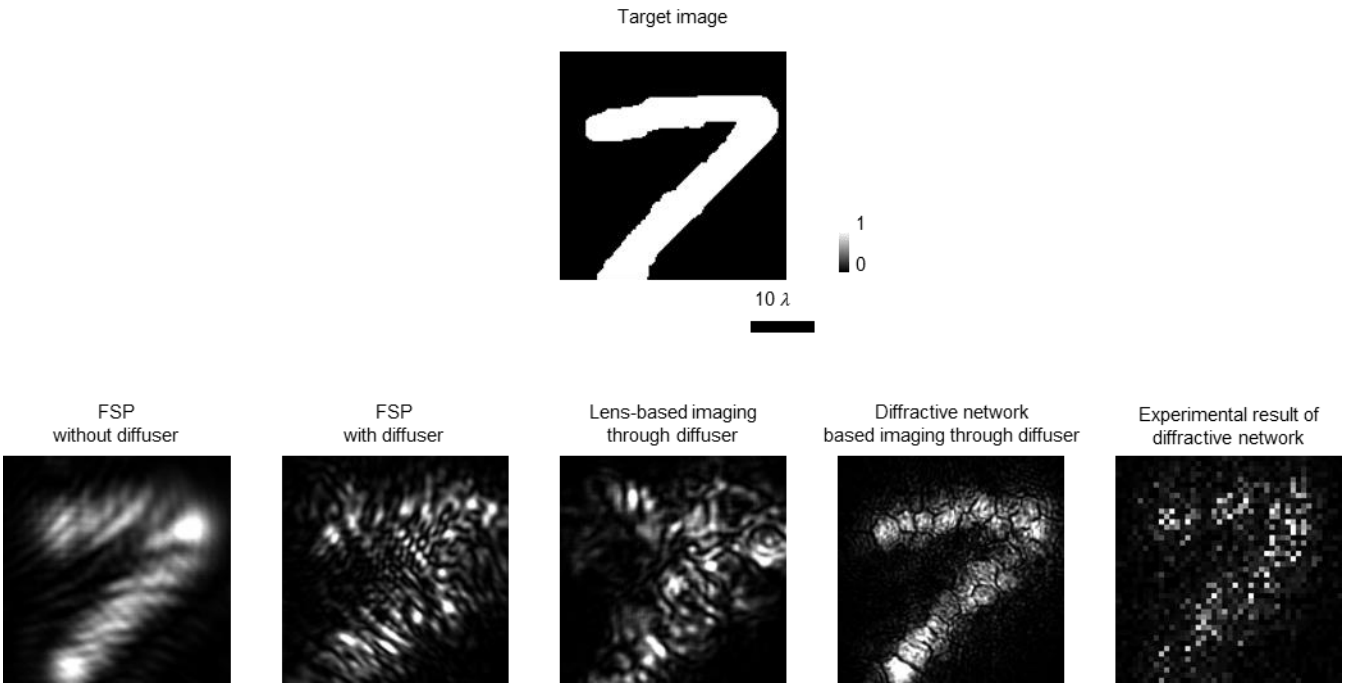


**Figure S2. Numerical and experimental results showing the image distortion generated by a random phase diffuser.** To exemplify the image distortion generated by a phase diffuser, the intensity distribution at the output field-of-view was numerically simulated as a result of the free space propagation (FSP) of the input object '7', without and with the presence of the diffuser K1 that was randomly generated (first and second panels in the second row, respectively); imaging of the same input object through the diffuser using an aberration-free lens is also shown in the third panel. All-optical reconstruction of the trained diffractive network and its experimental counterpart are also shown in the last two panels on the right. For better visualization, the contrast of all the panels is enhanced using the default MATLAB function (*imadjust*) (see the Methods section of the main text for details).

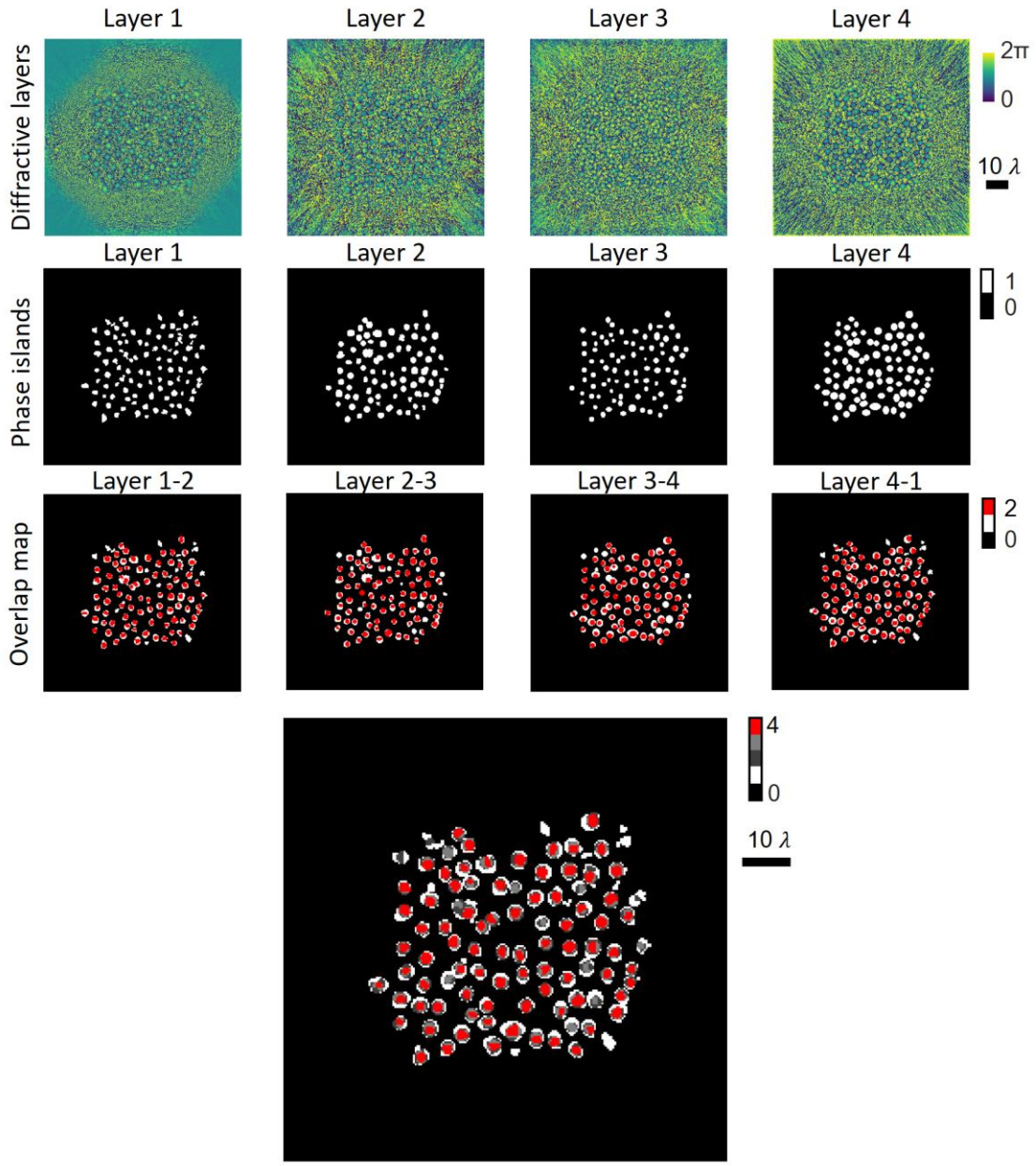


**Figure S3. Overlap map of phase islands on successive diffractive layers.** After training, the converged diffractive layers consist of multiple smooth phase islands, as shown in the first row. Layers corresponding to the diffractive network that was trained using n=20 diffusers at each epoch were used here as an example. Binary spatial maps were generated to mark the locations of these phase islands on each layer (second row). The summation of two consecutive spatial maps was used to illustrate the overlap between the phase islands on successive diffractive layers (as well as between phase islands on the first and last layers), as displayed in the third row with the overlapped regions marked in red. Similarly, an overlap map among all four diffractive layers is depicted in the last row.

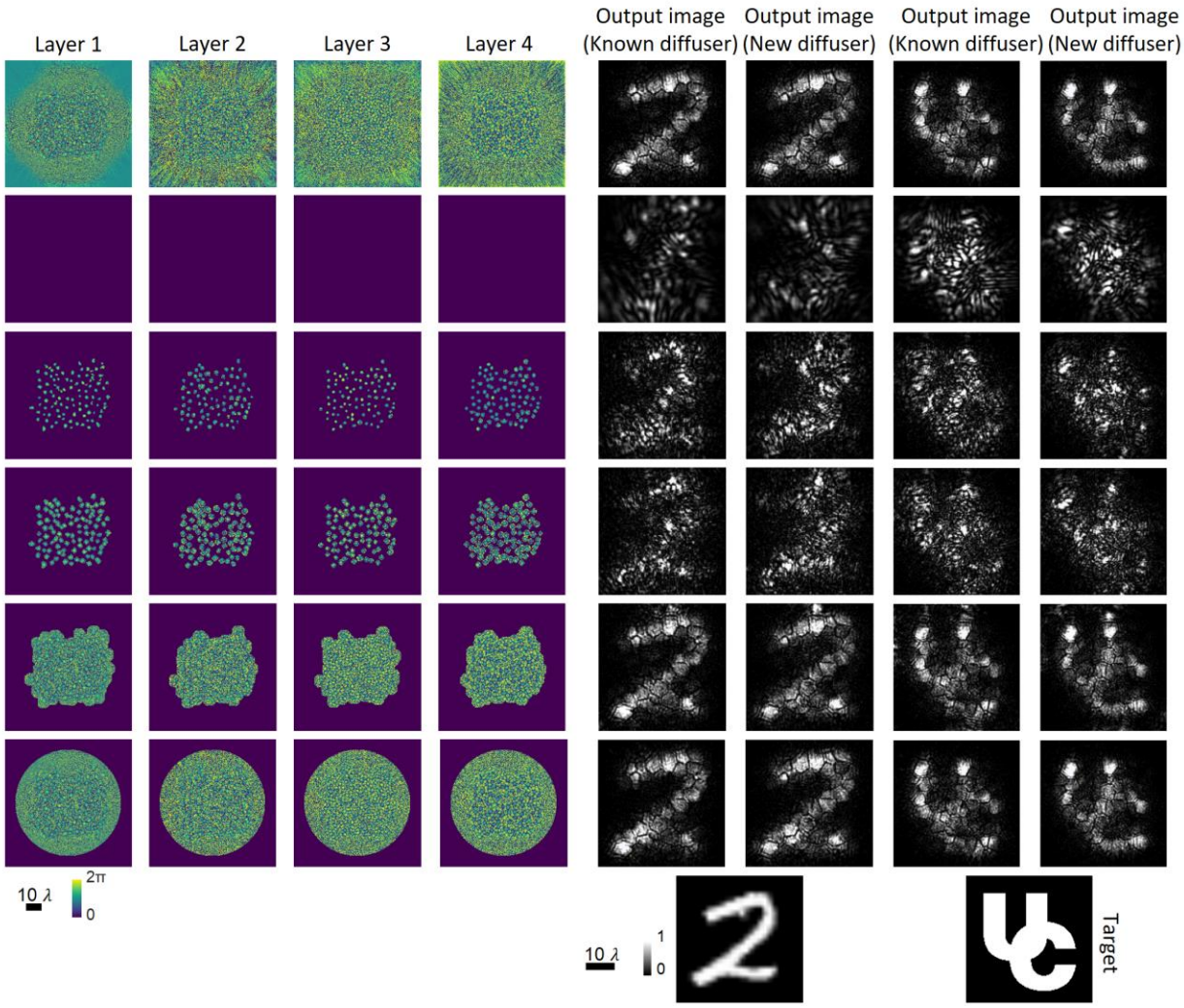


**Figure S4. Comparison of diffractive network output images under different levels of pruning.** Pruning of neurons outside the phase islands of each diffractive layer was conducted and the resulting diffractive layers were used to image through unknown random diffusers. The output images of the diffractive network that was trained using n=20 diffusers at each epoch are shown in the first row as a baseline. The PCC values of the output images of handwritten digit '2' distorted by a known and new diffuser were calculated to be 0.7797 and 0.7932, respectively. The intensity distribution of the output field-of-view without the presence of the diffractive layers is also shown (second row) and the corresponding PCC values were calculated to be 0.5321 and 0.5350 with a known and a new diffuser, respectively. The spatial maps presented in Figure S3 were used to prune the diffractive layers by dropping out the neurons outside the phase islands, i.e., only keeping the phase modulation provided by those phase islands and assuming that other neurons provide zero phase modulation. The pruned diffractive layers and the corresponding output images are shown in the third row, and the PCC values of the output images of handwritten digit '2' distorted by a known and new diffuser were calculated to be 0.5487 and 0.5452 with a known and a new diffuser, respectively. Further dilation of the binary spatial maps to include neurons right adjacent to the phase islands failed to significantly improve the reconstruction fidelity (fourth row), yielding PCC values of 0.5627/0.5344. Using all the neurons/pixels within the contour defined by the phase islands, the reconstructed images still contained some artifacts as shown in the fifth row, providing PCC values of 0.7543/0.7340. Finally, using all the pixels on each layer

within a circular aperture of $80\lambda$ diameter further improved the output images, yielding PCC values of 0.7797/0.7934. A more complicated input object and its reconstruction results are also presented in the right two columns. This type of object was not included in our training phase, which only involved handwritten digits (MNIST).

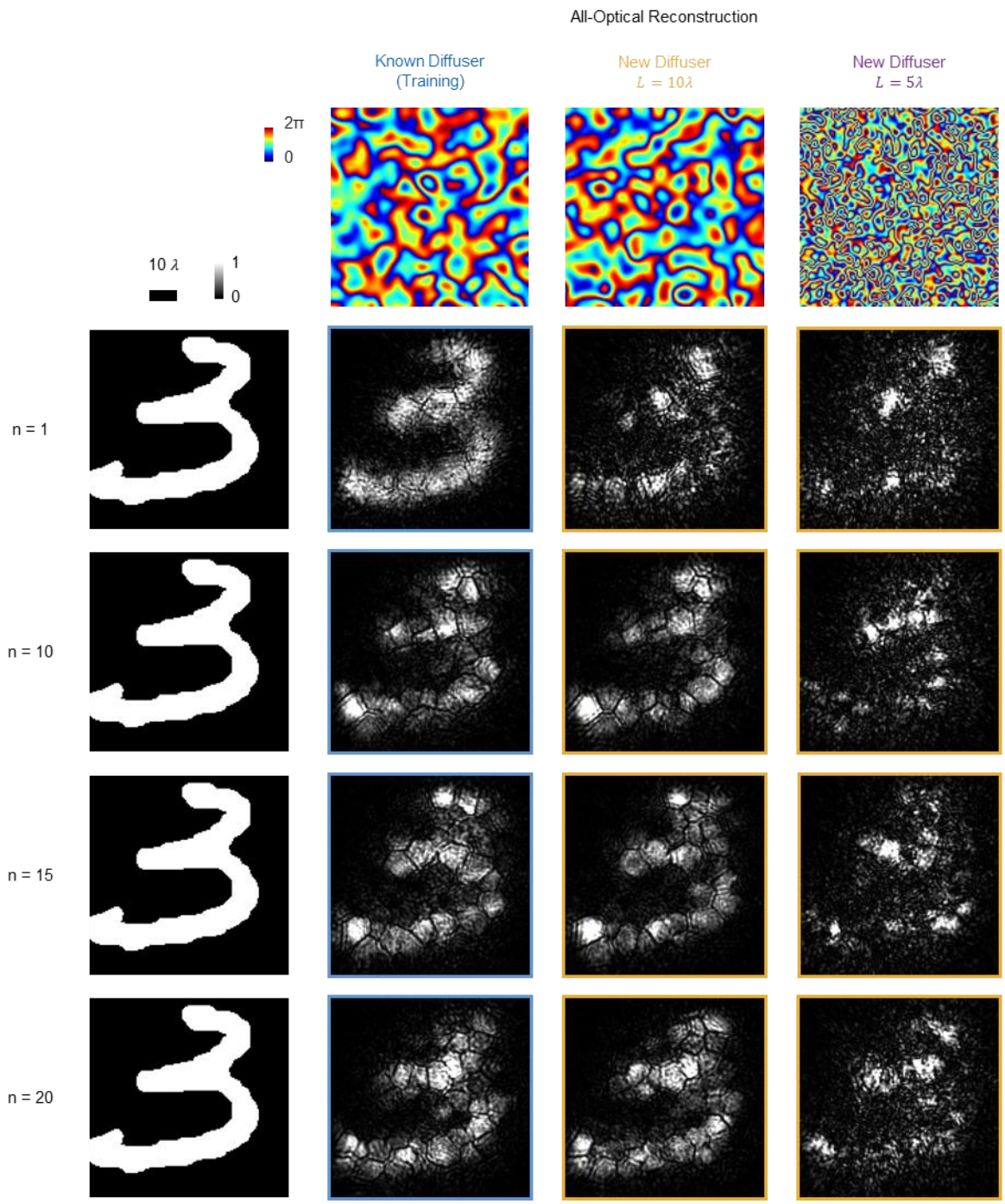


**Figure S5. Imaging through random diffusers with different correlation lengths ($L$).** Diffractive networks designed/trained with a larger correlation length ($L = 10\lambda$) were used to blindly reconstruct images distorted by an unknown, new diffuser with a smaller correlation length ($L = 5\lambda$). Although these diffractive networks were solely trained using random phase diffusers with $L = 10\lambda$, they were still successful in all-optical imaging through a random diffuser with $L = 5\lambda$; see e.g., the network that used $n$ = 20, the bottom row. For better

visualization, the contrast of the diffractive images is enhanced using the default MATLAB function (*imadjust*) (see the Methods section of the main text for details).